

Module Title: Mixed Models with Medical Applications**©UNIVERSITY OF LEEDS****School of Mathematics****Semester Two 202223****Calculator instructions:**

- You are allowed to use a non-programmable calculator in this examination.

Dictionary instructions:

- You are not allowed to use your own dictionary in this exam. A basic English dictionary is available to use. Raise your hand and ask an invigilator if you need it.

Exam information:

- There are 39 pages to this examination.
- There will be **2 hours 30 minutes** to complete this examination.
- This examination is worth 80% of the module mark
- Statistical tables are provided at the end of the question paper.
- Write your answers to each question in the allocated box below each question, otherwise it will not be marked (e.g. if you write your answer for question 1(b) in the box for question 2, it will not be marked).
- You do not need to fill the allocated answer box. Use as much or as little space as you need to answer the question.
- If you need more space for any of your answers, use the additional pages at the back of your booklet.
- You must write your name, Student ID Number and seat number in the relevant boxes on the question paper.
- There are six questions in this exam, each worth 25 marks. Please attempt all questions.
- Numbers in brackets indicate the marks available for individual parts of each question.
- You must show your working and state your answer clearly.

Please do not remove this paper from the exam venue.

NAME										
STUDENT NUMBER										
SEAT NUMBER										

1. (a) In general, for matrices $\mathbf{A}$ and $\mathbf{B}$, how is the Kronecker product $\mathbf{A} \otimes \mathbf{B}$ defined? [2 marks]

- (b) Explain each of the four steps in the following R code and write down the matrix **M** that it produces. **[5 marks]**

```
J <- rep(1, 9)
dim(J) <- c(3, 3)
M <- diag(6) + 25 * diag(2) %x% J
M <- M[1:5, 1:5]
```

- (c) How would you change the code in part (b) so that the matrix M produced would be [4 marks]

$$\mathbf{S} = \begin{pmatrix} 10 & 1 & 0 & 0 \\ 1 & 10 & 0 & 0 \\ 0 & 0 & 10 & 1 \\ 0 & 0 & 1 & 10 \end{pmatrix} ?$$

- (d) Let S be the matrix from part (c). The following code simulates a sample of [10 marks] observations from a random intercept model. Write this model down in clear mathematical notation and state the **numerical values** of all parameters in the model, including the variances at each level and the ranges of any indices. Also write down the design matrix.

```
library("MASS")
```

```
int <- c(1,1,1,1)
```

```
x1 <- c(1,1,0,0)
```

```
x2 <- c(0,1,0,1)
```

```
y <- mvrnorm(n = 1, mu = cbind(int, x1, x2) %*% c(1,-1,1), Sigma = S)
```

- (e) Use the covariance matrix from part (c) to calculate the intra-class correlation coefficient (ICC) for this model. What does the value of this ICC suggest about the samples our simulation will produce in part (d)? **[4 marks]**

2. Consider a (random) response vector $\mathbf{y}$ of length n and a $n \times (p + 1)$ design matrix $\mathbf{X}$. Suppose that our model for the distribution of $\mathbf{y}$ has the form

$$\mathbf{y} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\boldsymbol{\Omega}),$$

where $\boldsymbol{\Omega}$ is a positive definite $n \times n$ matrix, $\boldsymbol{\beta}$ is a vector of length $p + 1$ and σ^2 is a positive constant.

The method used to estimate the model's parameters will usually depend on what we are willing to assume about the matrix $\boldsymbol{\Omega}$.

- (a) Under what assumption about $\boldsymbol{\Omega}$ would we use the ordinary least squares (OLS) estimator [2 marks]

$$\hat{\boldsymbol{\beta}}_{OLS} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

to estimate $\boldsymbol{\beta}$?

- (b) Under what assumption about $\boldsymbol{\Omega}$ might we use the generalised least squares estimator [2 marks]

$$\hat{\boldsymbol{\beta}}_{GLS} = (\mathbf{X}^T \boldsymbol{\Omega}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\Omega}^{-1} \mathbf{y}$$

to estimate $\boldsymbol{\beta}$, instead of the OLS estimator?

- (c) By the Cholesky decomposition, we know there exists an invertible matrix $\mathbf{S}$ such that $\mathbf{\Omega} = \mathbf{S}\mathbf{S}^T$. Prove that **[10 marks]**

$$\mathbf{S}^{-1}\mathbf{y} \sim N((\mathbf{S}^{-1}\mathbf{X})\boldsymbol{\beta}, \sigma^2\mathbf{I}_n).$$

State any general theorem that you use in this proof.

- (d) Explain why we can apply the OLS estimator from part (a) to the result stated in part (c). Show that doing this gives the GLS estimator from part (b). **[8 marks]**

- (e) Explain briefly why the GLS estimator might not be suitable when estimating the parameters of a multilevel model. Name at least one alternative method of estimation and explain why this alternative would be more suitable than just GLS? **[3 marks]**

3. Consider the multilevel model

$$y_{ij} = (\beta_0 + u_{0j}) + (\beta_1 + u_{1j})x_{1ij} + (\beta_2 + u_{2j})x_{2ij} + e_{0ij},$$

$$e_{0ij} \stackrel{iid}{\sim} N(0, \sigma_{e0}^2), \quad u_{0j} \stackrel{iid}{\sim} N(0, \sigma_{u0}^2), \quad u_{1j} \stackrel{iid}{\sim} N(0, \sigma_{u1}^2), \quad u_{2j} \stackrel{iid}{\sim} N(0, \sigma_{u2}^2),$$

where $i = 1, 2, \dots, n_j$ and $j = 1, \dots, m$. Assume that all random terms on the right hand side of this model (u_{0j} , u_{1j} , u_{2j} and e_{0ij}) are mutually independent of each other, except for a possible covariance $\sigma_{u12} = \text{cov}(u_{1j}, u_{2j})$.

- (a) When compared to a random intercept model with the same response variable and the same covariates, how many additional unknown parameters does this model (stated above) have? **[2 marks]**

- (b) Which hypothesis test could be performed to compare this model to a random intercept model? State the name of the test and state the null hypothesis tested. **[3 marks]**

- (c) Suppose that we want **unbiased** estimates of the parameters in our model. Suppose that we plan to obtain such estimates by running the R code **[8 marks]**

```
library("MASS")
```

```
lmer(y = 1 + x1 + (x1 + x2 | group), REML = FALSE)
```

where y , x_1 and x_2 will be vectors containing our observations of the variables y , x_1 and x_2 respectively, and where $group$ will be a vector describing which cluster (j) each observation belongs to. Point out any mistakes you find in this code and how you would correct each of them.

- (d) Assume that there are at least two observations in cluster j . Derive a formula for the covariance between y_{1j} and y_{2j} . **[9 marks]**

- (e) For what values of j and k would y_{ij} and y_{hk} have zero covariance? Explain your answer. **[3 marks]**

4. Suppose that data have been collected on thirty packs (groups) of wolves. Each pack lived in one of eight distinct locations in North America. In total there are two-hundred wolves in the sample. Our data set contains information on each of these wolves, including which pack and which location each wolf belonged to.

The response variable (y) that we would like to analyse is the life span (time from birth until death, measured in years) of these wolves. There are also two covariates in the data set:

$$x_1 = \begin{cases} 1 & \text{male wolf} \\ 0 & \text{female wolf} \end{cases} \quad \text{and} \quad x_2 = \begin{cases} 1 & \text{grey wolf} \\ 0 & \text{red wolf} \end{cases}.$$

- (a) Multilevel models assume the data to have a hierarchical structure. In this data set, we have a possible three-level hierarchical structure, with wolves grouped by pack and pack grouped by location. Explain what “hierarchical” means here, both in words and with a small sketch. **[3 marks]**

- (b) Give a single example of how our data could have a **non-hierarchical structure**. [4 marks]
Is this an example of multiple membership or cross classification? Explain this structure in words and with a small sketch.

(c) Assume that the data **are hierarchical**. Write down **just one** equation for a multilevel model that would potentially allow us to answer each of the following questions about the life span of wolves: **[7 marks]**

- How much does it vary between wolves?
- How much does it vary between packs?
- How much does it vary between locations?
- How much is it affected by being grey?
- How much is it affected by being male?
- Does the effect of being male vary between locations?

State the meaning of any variables and any indices in the model clearly, but there is no need to state any assumptions about distributions or independence yet.

- (d) Write down the **random part** of your model from part (c). If the response variable (life span) is assumed to be normal (given the covariate data), state any assumptions that you would make about probability distributions or independence for the random terms on the right hand side of your model. **[5 marks]**

- (e) Write down the **fixed part** of your model. In terms of the parameters in this fixed part, write down a single null hypothesis that could be used for simultaneously testing the following two statements: **[4 marks]**
- The average life span of red, female wolves is 5 years.
 - The combined effect of being male and grey on lifespan is an increase of 1 year.

Express your null hypothesis using a contrast matrix.

- (f) Explain at least two ways in which shrunken residuals calculated from your model could potentially be used in this analysis. **[2 marks]**

5. A cluster randomised trial has been proposed to compare a new intervention, aiming to improve a continuous measure of mental health for patients with depression, against the current standard of care. The intervention is delivered by general practitioners (GPs), with these GPs being randomly allocated to the intervention or the control treatment.

An initial sample size calculation which ignored the potential clustering effect of GPs used the formula

$$1 - \beta = 1 - \Phi \left(\Phi^{-1}(1 - \alpha) - \frac{\delta_1 - \delta_0}{\sqrt{2\sigma^2/N}} \right)$$

(where $\Phi(\cdot)$ denotes the standard normal distribution function) to find that $N = 175$ patients in each arm would be needed to provide a power of $1 - \beta = 0.8$ to detect a Minimal Clinically Important Difference of $\delta_1 = 0.3$ when using a one-sided test with type I error rate $\alpha = 0.025$ to test the null hypothesis of $\delta_0 = 0$. The between-GP variance is thought to equal $\sigma_u^2 = 0.05$, and the total variance $\sigma^2 = 1$. Denote the outcome of the i -th patient in the j -th cluster of arm k by $y_{i,j,k}$.

Three GPs were recruited to each arm of the study. In the control arm, the GPs recruited $n_{1,0} = 40$, $n_{2,0} = 60$ and $n_{3,0} = 75$ patients; in the intervention arm, they recruited $n_{1,1} = 40$, $n_{2,1} = 60$, and $n_{3,1} = 75$. The two arms will be compared using their weighted sample means

$$\tilde{y}_k = \frac{\sum_{j=1}^m n_{j,k} \left(\frac{1}{n_{j,k}} \sum_{i=1}^{n_{j,k}} y_{i,j,k} \right)}{\sum_{j=1}^m n_{j,k}}.$$

- (a) Based on the initial sample size calculation and using the fact that $\Phi^{-1}(0.2) = -0.84$, what is the total variance in the outcome measure? What is the correlation between two patients with the same GP?

[5 marks]

- (b) Using the design effect for cluster randomised trials with cluster size $n_{j,k}$ in the j -th cluster of the k -th arm,

$$1 + \left(\frac{\sum_{j=1}^m n_{j,k}^2}{\sum_{j=1}^m n_{j,k}} - 1 \right) \rho$$

(where ρ denotes the Intraclass Correlation Coefficient), what is the actual power of the trial?

[4 marks]

- (c) To what extent could power be improved if the patients were allocated more equally amongst GPs?

[3 marks]

- (d) Using the given design effect or otherwise, derive an expression for the variance of the weighted sample mean $\tilde{y}_k$ in terms of the between-GP variance σ_u^2 and within-GP variance σ_e^2 .

[6 marks]

- (e) Does the weighted or unweighted estimator (where the latter is defined as $\bar{y}_k = \frac{1}{m} \sum_{j=1}^m \left(\frac{1}{n_{j,k}} \sum_{i=1}^{n_{j,k}} y_{i,j,k} \right)$) give lower variance in the extreme case where the Intraclass Correlation Coefficient $\rho = 1$? Justify your answer.

[7 marks]

(You may want to use that

$$\left(\sum_{j=1}^m n_j \right)^2 \leq m \sum_{j=1}^m n_j^2$$

by the Cauchy-Schwarz inequality).

6. Consider a clinical trial evaluating a new talking therapy intervention, delivered to patients by a trained therapist over the course of six treatment sessions. It has been suggested that patients should be allowed to receive their treatment from any one of the six therapists taking part in the trial for any of their treatment sessions, and that the clustering effects due to therapists will be additive and in proportion to the number of treatment sessions they deliver for the patient in question. The control treatment does not have any clustering effects associated with it.

- (a) Write down an appropriate mixed model for the outcome of the i -th participant in arm k of the trial, y_{ik} , and state any assumptions.

[4 marks]

- (b) Derive expressions for the variance of $y_{i,k}$ and the correlation between $y_{i,k}$ and $y_{i',k}$ in the intervention arm of the trial.

[7 marks]

(c) The first three rows of the data are

#	y	trt	d1	d2	d3	d4	d5	d6
# 1	3.37278338	1	1	6	3	2	2	4
# 2	2.28561021	1	5	4	4	4	4	3
# 3	3.10059629	1	4	4	2	5	2	2

with the variable d_s denoting the ID of the therapist who delivered treatment sessions s for that patient. If the between-therapist variance is $\sigma_u^2 = 2$, determine the covariance between patients 2 and 3.

[5 marks]

- (d) Using data of the given form, write down the formula component of the brms code you would run to estimate the treatment effect. Suggest a criteria which could be used to decide if the treatment should be adopted in practice.

[5 marks]

- (e) Suppose we can control exactly which therapists deliver which treatment sessions for any given patient. What allocation scheme would minimise the variance of that patient's outcome? Justify your answer.

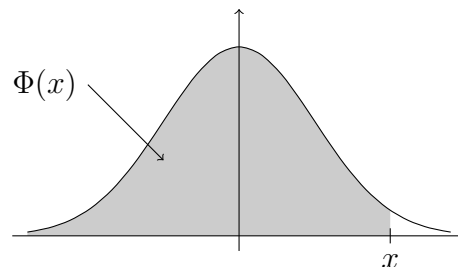
[4 marks]

Normal Distribution Function Tables

The first table gives

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{1}{2}t^2} dt$$

and this corresponds to the shaded area in the figure to the right. $\Phi(x)$ is the probability that a random variable, normally distributed with zero mean and unit variance, will be less than or equal to x . When $x < 0$ use $\Phi(x) = 1 - \Phi(-x)$, as the normal distribution with mean zero is symmetric about zero. To interpolate, use the formula



$$\Phi(x) \approx \Phi(x_1) + \frac{x - x_1}{x_2 - x_1} (\Phi(x_2) - \Phi(x_1))$$

Table 1

x	$\Phi(x)$	x	$\Phi(x)$	x	$\Phi(x)$	x	$\Phi(x)$	x	$\Phi(x)$	x	$\Phi(x)$
0.00	0.5000	0.50	0.6915	1.00	0.8413	1.50	0.9332	2.00	0.9772	2.50	0.9938
0.05	0.5199	0.55	0.7088	1.05	0.8531	1.55	0.9394	2.05	0.9798	2.55	0.9946
0.10	0.5398	0.60	0.7257	1.10	0.8643	1.60	0.9452	2.10	0.9821	2.60	0.9953
0.15	0.5596	0.65	0.7422	1.15	0.8749	1.65	0.9505	2.15	0.9842	2.65	0.9960
0.20	0.5793	0.70	0.7580	1.20	0.8849	1.70	0.9554	2.20	0.9861	2.70	0.9965
0.25	0.5987	0.75	0.7734	1.25	0.8944	1.75	0.9599	2.25	0.9878	2.75	0.9970
0.30	0.6179	0.80	0.7881	1.30	0.9032	1.80	0.9641	2.30	0.9893	2.80	0.9974
0.35	0.6368	0.85	0.8023	1.35	0.9115	1.85	0.9678	2.35	0.9906	2.85	0.9978
0.40	0.6554	0.90	0.8159	1.40	0.9192	1.90	0.9713	2.40	0.9918	2.90	0.9981
0.45	0.6736	0.95	0.8289	1.45	0.9265	1.95	0.9744	2.45	0.9929	2.95	0.9984
0.50	0.6915	1.00	0.8413	1.50	0.9332	2.00	0.9772	2.50	0.9938	3.00	0.9987

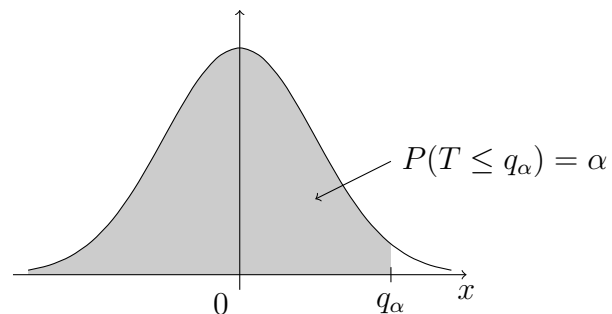
The inverse function $\Phi^{-1}(p)$ is tabulated below for various values of p .

Table 2

p	0.900	0.950	0.975	0.990	0.995	0.999	0.9995
$\Phi^{-1}(p)$	1.2816	1.6449	1.9600	2.3263	2.5758	3.0902	3.2905

Quantiles of the t -Distribution

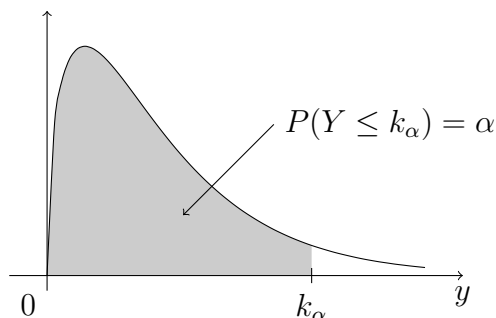
This table gives the α -quantiles for the $t(\nu)$ -distribution with various degrees of freedom ν , and for various values of α , as indicated by the figure to the right. Quantiles for the lower tail can be found using the symmetry $P(T \leq -q) = P(T \geq q) = 1 - P(T \leq q)$. The limiting distribution of $t(\nu)$ as $\nu \rightarrow \infty$ is the standard normal distribution.



ν	$\alpha = 0.9$	$\alpha = 0.95$	$\alpha = 0.975$	$\alpha = 0.99$	$\alpha = 0.995$
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
11	1.363	1.796	2.201	2.718	3.106
12	1.356	1.782	2.179	2.681	3.055
13	1.350	1.771	2.160	2.650	3.012
14	1.345	1.761	2.145	2.624	2.977
15	1.341	1.753	2.131	2.602	2.947
16	1.337	1.746	2.120	2.583	2.921
17	1.333	1.740	2.110	2.567	2.898
18	1.330	1.734	2.101	2.552	2.878
19	1.328	1.729	2.093	2.539	2.861
20	1.325	1.725	2.086	2.528	2.845
25	1.316	1.708	2.060	2.485	2.787
30	1.310	1.697	2.042	2.457	2.750
35	1.306	1.690	2.030	2.438	2.724
40	1.303	1.684	2.021	2.423	2.704
45	1.301	1.679	2.014	2.412	2.690
50	1.299	1.676	2.009	2.403	2.678
60	1.296	1.671	2.000	2.390	2.660
80	1.292	1.664	1.990	2.374	2.639
100	1.290	1.660	1.984	2.364	2.626

Quantiles of the χ^2 -Distribution

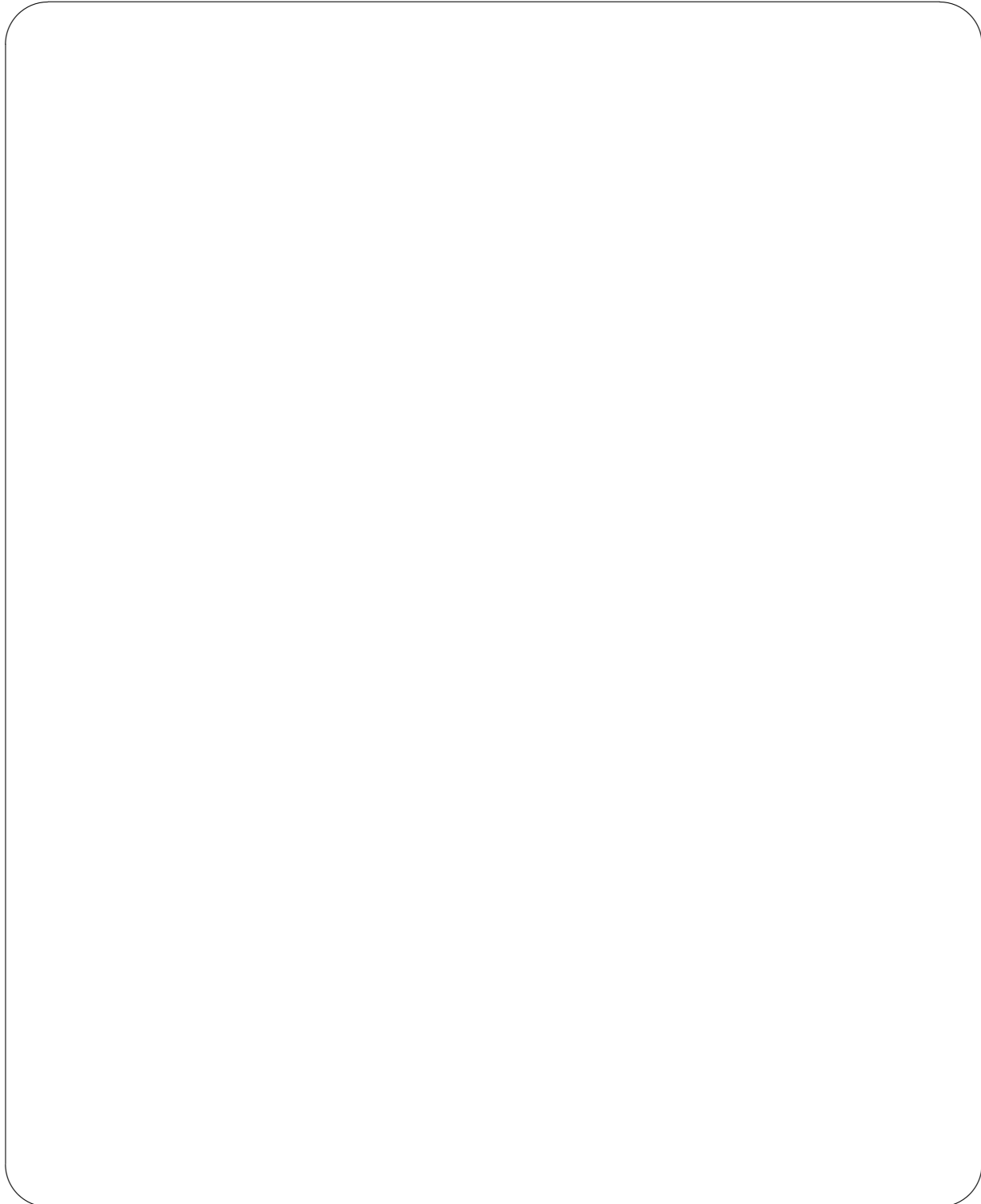
This table gives the α -quantiles for the χ^2_ν -distribution with various degrees of freedom ν , and for various values of α , as indicated by the figure to the right. (The figure uses $\nu = 3$.) If $Y \sim \chi^2_\nu$ for $\nu > 100$, then $\sqrt{2Y}$ is approximately normally distributed with mean $\sqrt{2\nu - 1}$ and unit variance.



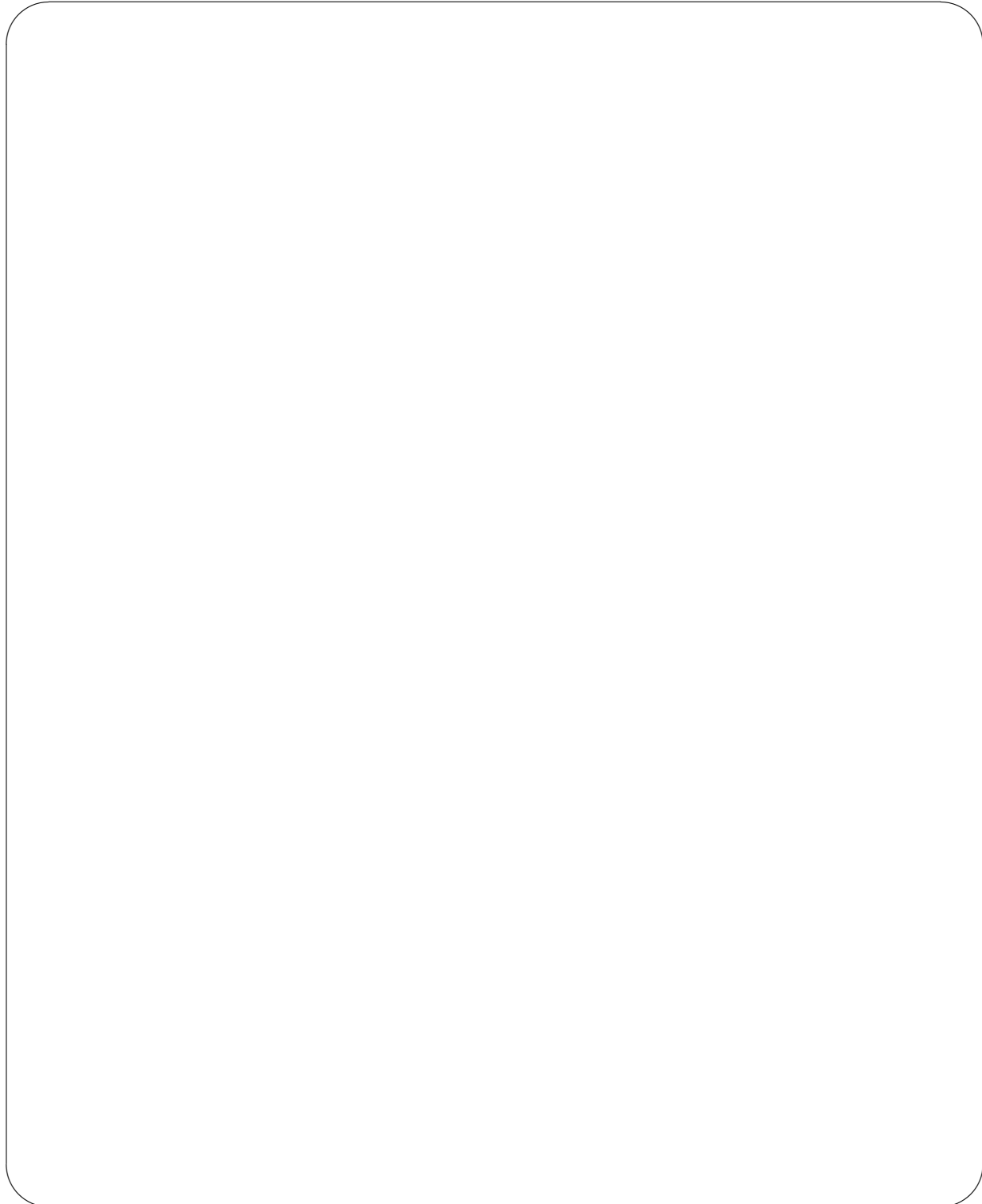
ν	$\alpha = 0.025$	$\alpha = 0.9$	$\alpha = 0.95$	$\alpha = 0.975$	$\alpha = 0.99$	$\alpha = 0.995$
1	0.001	2.706	3.841	5.024	6.635	7.879
2	0.051	4.605	5.991	7.378	9.210	10.597
3	0.216	6.251	7.815	9.348	11.345	12.838
4	0.484	7.779	9.488	11.143	13.277	14.860
5	0.831	9.236	11.070	12.833	15.086	16.750
6	1.237	10.645	12.592	14.449	16.812	18.548
7	1.690	12.017	14.067	16.013	18.475	20.278
8	2.180	13.362	15.507	17.535	20.090	21.955
9	2.700	14.684	16.919	19.023	21.666	23.589
10	3.247	15.987	18.307	20.483	23.209	25.188
11	3.816	17.275	19.675	21.920	24.725	26.757
12	4.404	18.549	21.026	23.337	26.217	28.300
13	5.009	19.812	22.362	24.736	27.688	29.819
14	5.629	21.064	23.685	26.119	29.141	31.319
15	6.262	22.307	24.996	27.488	30.578	32.801
16	6.908	23.542	26.296	28.845	32.000	34.267
17	7.564	24.769	27.587	30.191	33.409	35.718
18	8.231	25.989	28.869	31.526	34.805	37.156
19	8.907	27.204	30.144	32.852	36.191	38.582
20	9.591	28.412	31.410	34.170	37.566	39.997
25	13.120	34.382	37.652	40.646	44.314	46.928
30	16.791	40.256	43.773	46.979	50.892	53.672
35	20.569	46.059	49.802	53.203	57.342	60.275
40	24.433	51.805	55.758	59.342	63.691	66.766
45	28.366	57.505	61.656	65.410	69.957	73.166
50	32.357	63.167	67.505	71.420	76.154	79.490
60	40.482	74.397	79.082	83.298	88.379	91.952
80	57.153	96.578	101.879	106.629	112.329	116.321
100	74.222	118.498	124.342	129.561	135.807	140.169

This is the end of the examination paper. The remaining pages are for you to use if you run out of space in the boxes provided for each answer.

- This additional page may be used if you ran out of space answering any question in the box provided for that question.
- Write the question number and continue your answer in the box on this page.
- You may continue more than one answer in the boxes on this page, but you must indicate clearly where each continued answer starts and which question is being answered.

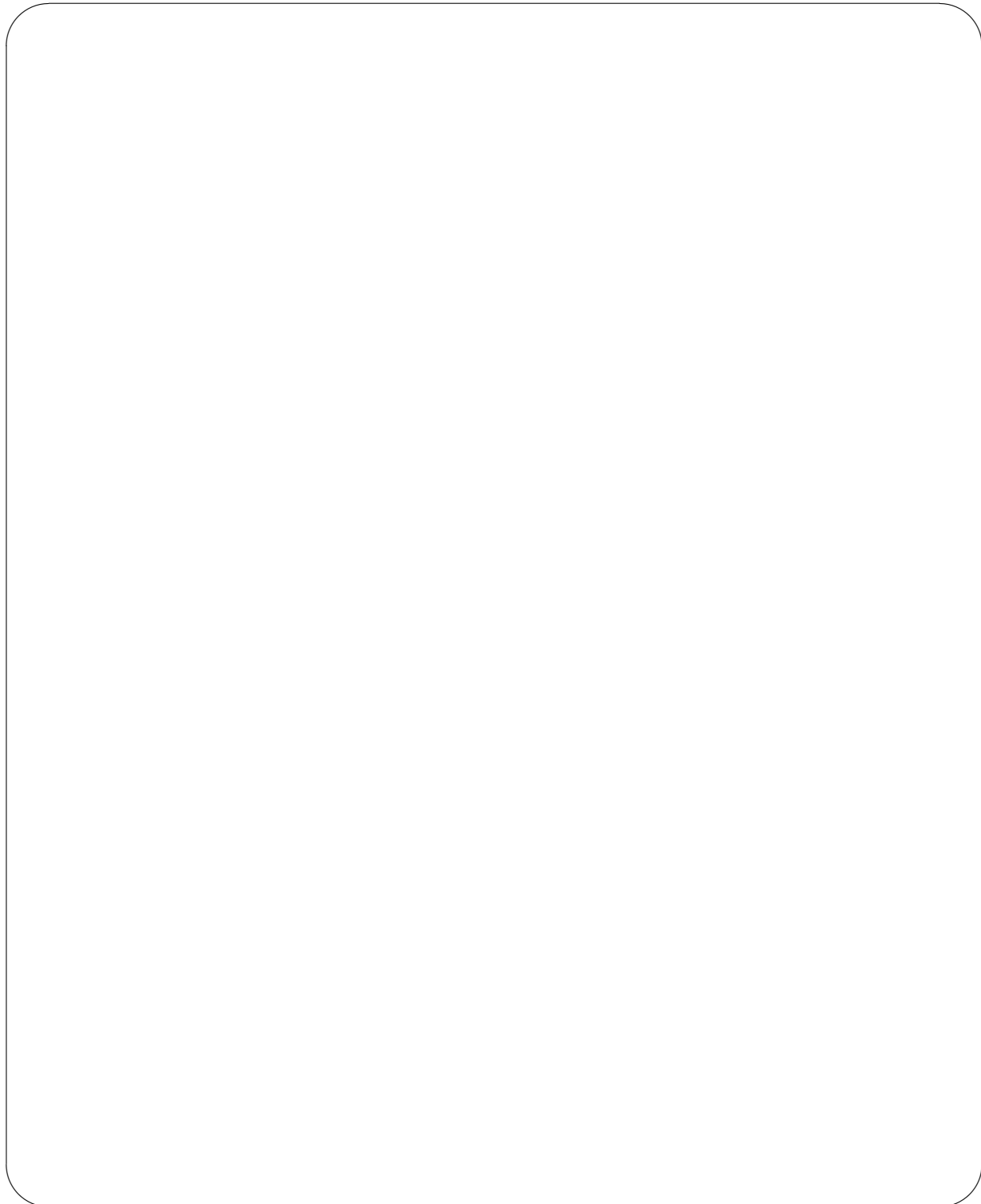


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